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Perspective

# Yield Smarter, Not Harder: Good Practices for Machine Learning of Reaction Outcomes

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## Challenges in Reaction Yield Prediction

### 1. Descriptor complexity and computational cost

Complex descriptors incur substantial computational costs, while the benefit of greater complexity remains insufficiently tested.

### 2. Interpretability and usability

Deep-learning models often operate as black boxes, limiting interpretability and usability in synthetic settings.

### 3. Generalization beyond random splits

Random splits may share reactants or products across training and test sets, and often provide an upper bound on model performance.

### 4. Asymmetric yield distributions

Patent and literature data favor high yields, whereas HTE data contain more low-yield or failed reactions.

### 5. Missing or inconsistent metadata

Incomplete stoichiometry, reaction conditions and reaction times complicate modeling across data sets.

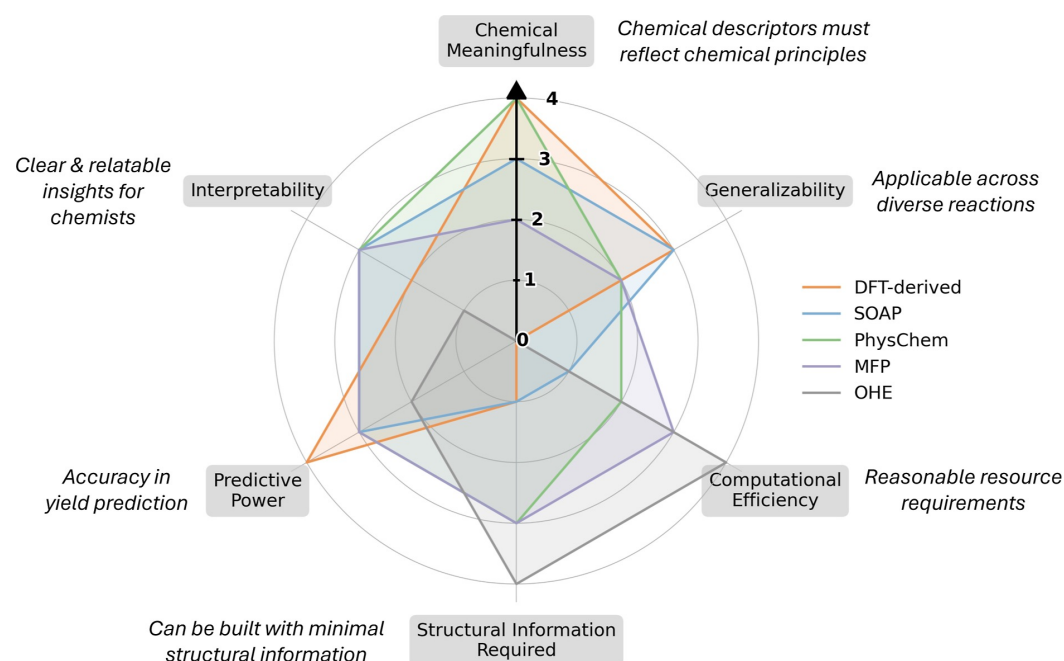


Figure 1. Qualitative comparison of descriptor types evaluated across six criteria: chemical meaningfulness, computational efficiency, amount of structural information required, generalizability, predictive power, and interpretability. Scores are given on a scale from 0 to 4, where higher values are more favorable. The five descriptor types ranked are density functional theory (DFT) derived, smooth overlap of atomic positions (SOAP),<sup>22</sup> physicochemical properties (PhysChem),<sup>23</sup> Morgan fingerprints (MFPs),<sup>24</sup> and one-hot encoding (OHE).<sup>25</sup>

### DFT: Density functional theory

Quantum-chemical features: orbital energies, atomic charges and reactivity indices.

### SOAP: Smooth overlap of atomic positions

3D atomic environments capturing steric and conformational features.

### PhysChem: Physicochemical properties

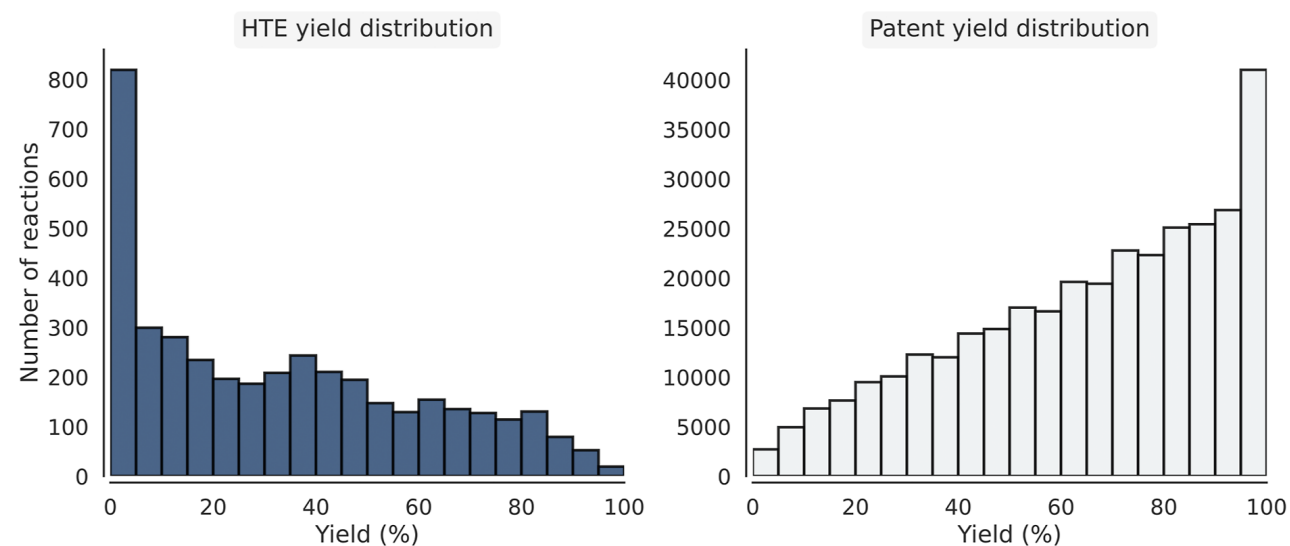
2D properties: size, polarity, hydrogen bonding and topological structure.

### MFP: Morgan fingerprints

Circular atom-centered substructures that encode molecular topology.

### OHE: One-hot encoding

Categorical identifiers for reaction components. Learns from identity without chemical structure.



### HTE data

Systematic screening retains many low-yield and failed reactions.

### Patent data

Publication bias favors successful, high-yielding reactions.

### Modeling challenge

Asymmetric yield distributions complicate training and evaluation.

Figure 2. Yield distributions from high-throughput experimentation (HTE, left) and USPTO patents (right). The HTE Buchwald–Hartwig data set of Ahneman et al.<sup>7</sup> includes a high proportion of low-yield or failed reactions, reflecting the exploratory nature of early-stage development. In contrast, the distribution in the USPTO data set of the same reaction class<sup>27</sup> is shifted toward high yields, illustrating a publication bias toward successful outcomes.

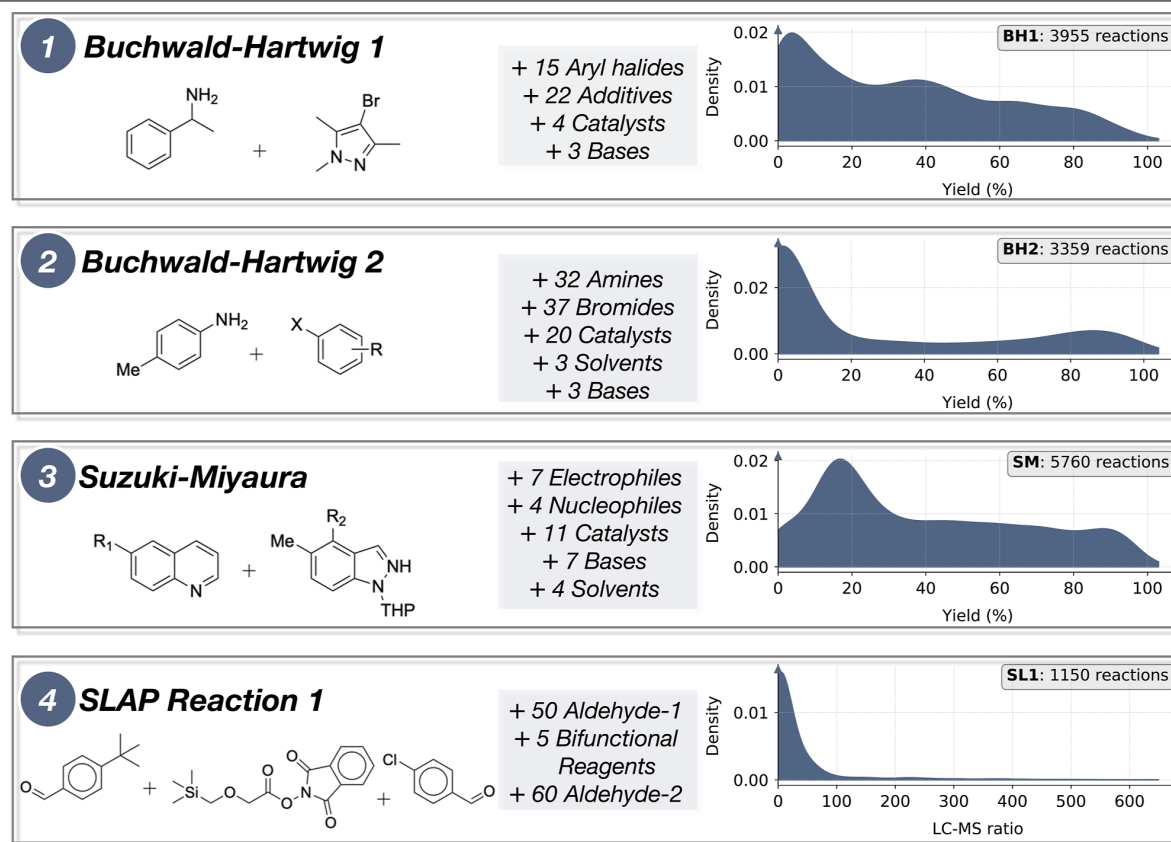
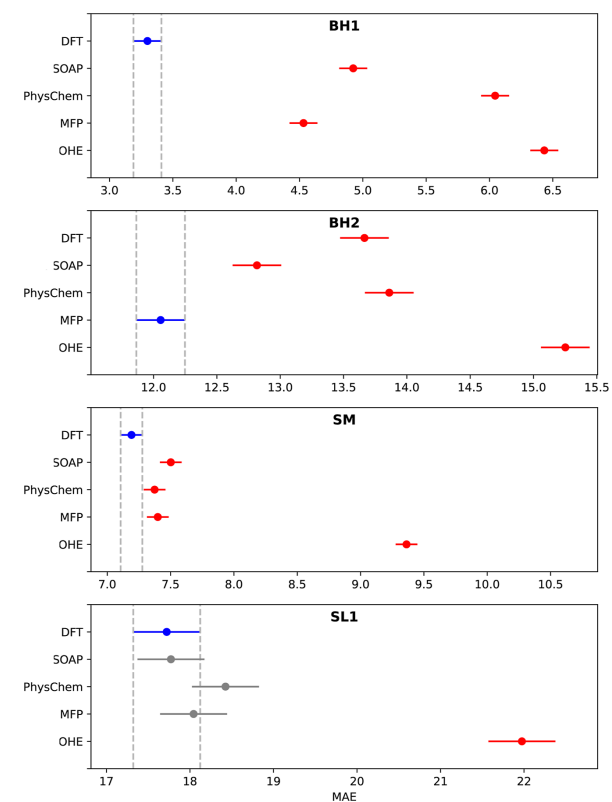
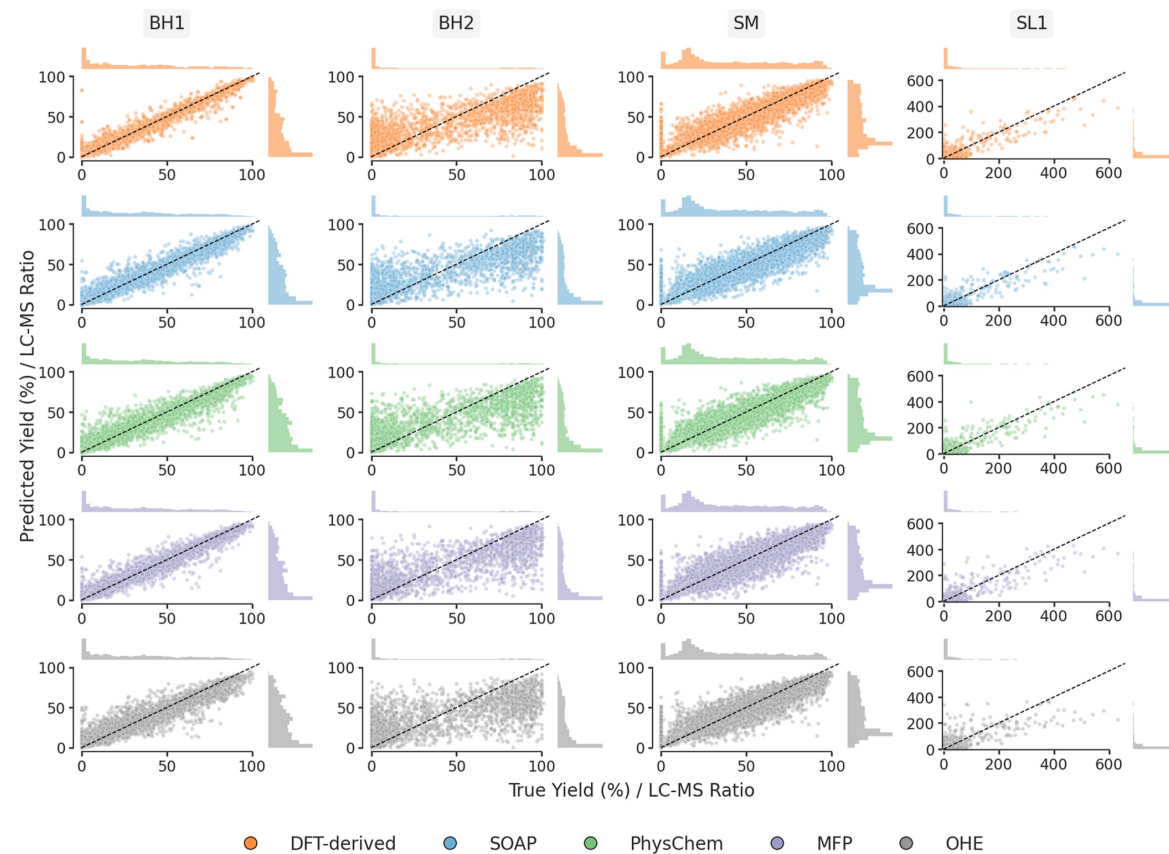
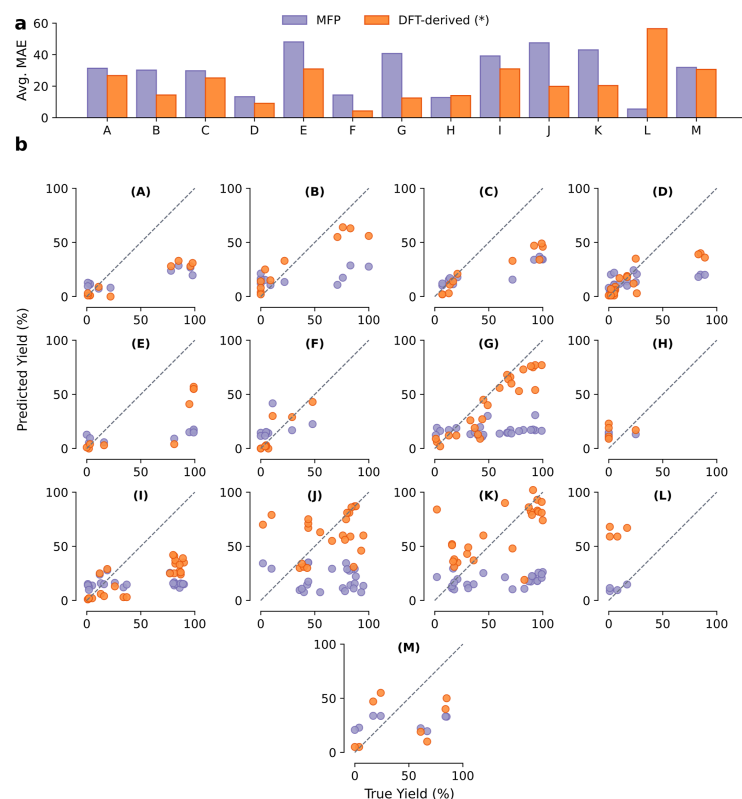


Figure 3. Overview of the four HTE data sets considered in this study, along with statistics on their reaction components: Buchwald–Hartwig amination (BH1 and BH2), Suzuki–Miyaura coupling (SM), and the SLAP reaction (SL1). Kernel density estimates of the observed yield values within each data set are shown on the right.





## External validation: 187 reactions

BH2 test reactions span 13 product groups (A–M).

## High-yield reactions remain difficult

Both models often fail to predict high yields.  
MFP predicts <50% for every external reaction.

## Interpret MAE with the yield distribution

Low MAE can reflect many low-yield reactions.  
MFP remains competitive at minimal cost.

Figure 6. Prediction performance on the BH2 external validation set with 187 out-of-sample reactions across product groups A–M for the RF regressors trained on MFP descriptors (purple) and for the ML model from Rinehart et al.<sup>8</sup> (orange), which was trained on DFT-derived descriptors (predicted yields directly taken from ref 8). (\*): Note that these DFT-derived descriptors are different from those in ref 7 that were used in the rest of this study. (a) Mean absolute error (MAE) per product group. For MFP, the model was also trained on the complete training set (in contrast to the cross-validation in Figure 4). (b) Parity plots of predicted versus experimental yields per product group.



## Conclusions and Recommendations

Reaction-yield prediction is not a “solved problem.”

Simple 2D-based models can rival more complex approaches in overall accuracy.

### 1. Avoid unnecessary descriptor complexity and include simple baselines.

Simpler models and descriptors can provide strong baselines and should always be included in a study for comparison.

### 2. Benchmark meaningfully.

Use evaluation protocols that reflect real-world application scenarios, such as component-wise data splits and external validation data sets.

### 3. Acknowledge and, if possible, address data imbalance.

Report performance across the yield distribution rather than relying solely on overall averages. Distributional skew remains an unsolved challenge.

### Reaction yield is not a purely structural property.

Progress requires more complete and consistently captured metadata on reaction conditions, stoichiometry and competitive pathways.

**Greater complexity needs statistically significant and practically relevant improvement.**



## S2.2 Descriptor Calculation

### S2.2.1 DFT-Derived Descriptors

BH1: descriptors from the original study, using Spartan at B3LYP/6-31G\*.

BH2, SM and SL1: ORCA 6.1, B3LYP/def2-TZVP. 33 descriptors per molecule: HOMO/LUMO energies and derived reactivity indices, frequencies and IR intensities of the first 10 normal modes, and CHELPG/Mulliken charge statistics.

### S2.2.2 SOAP

~500 conformers per molecule using RDKit srETKDGv3. RMSD pruning: 0.5 Å. MMFF94 or UFF optimization. Select the lowest-energy conformer. Cutoff: 6.0 Å. Six radial basis functions, spherical harmonics up to degree 4,  $\sigma = 0.5$ . Average across atoms and apply mu2 compression.

### S2.2.3 2D Physicochemical Descriptors

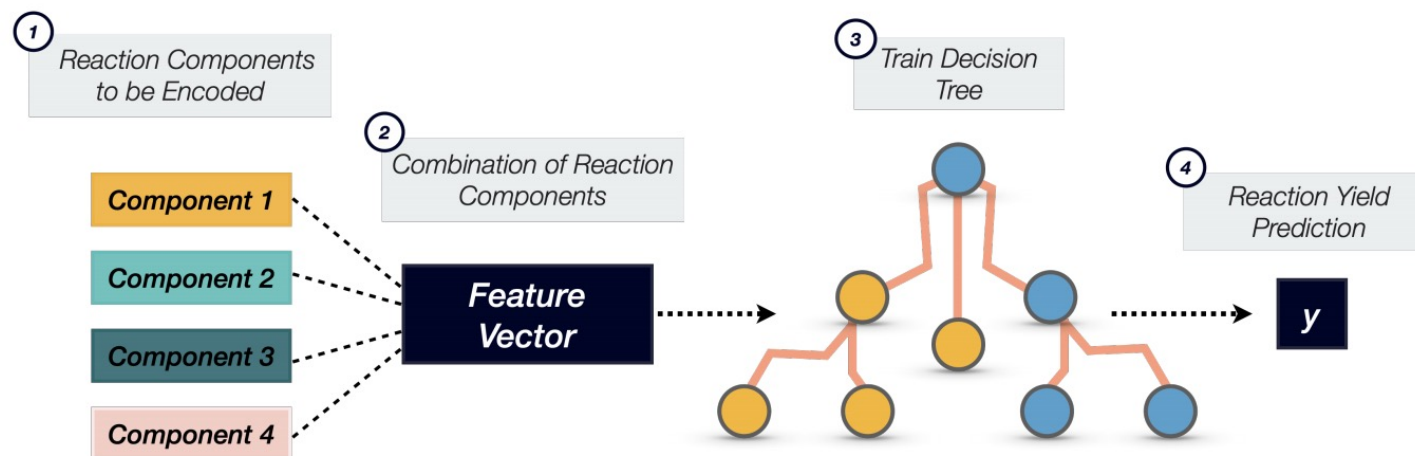
RDKit: 217 descriptors for each component type. Retain descriptors with no NaN and non-zero variance. Apply PCA to z-scaled descriptors. Retain enough components to explain 90% of variance.

### S2.2.4 Count-Based Morgan Fingerprints

RDKit count-based Morgan fingerprints. Radius of 3 and vector length of 1024. Preserve information about substructure frequency.

### S2.2.5 One Hot Encoding

Assign a categorical identifier to each reaction component. Structure-agnostic: learns patterns purely from identity. Encode after the train–test split and fit on the training set. Implementation: scikit-learn OneHotEncoder.



**Figure S1:** Workflow used in this study for reaction yield prediction using a RF regression model as ML algorithm: (1) descriptor calculation of the individual reaction components, (2) combination of the descriptors of the components into a feature vector, (3) training of the RF regression model, and (4) prediction of the reaction yield for a new instance.



**Table S3:** Mean absolute error (MAE), root-mean-square-error (RMSE), correlation coefficient  $R^2$ , and Kendall  $\tau$  of the predicted yield for the RF regression models trained with each descriptor across the four datasets (5x5 cross-validation). The standard deviation is given with  $\pm$ . For BH1, BH2, and SM, the MAE and RMSE values are in % (yields), whereas for SL1, the values reflect LC–MS product ratios.

| Dataset | Descriptor  | MAE            | RMSE           | $R^2$           | Kendall $\tau$  |
|---------|-------------|----------------|----------------|-----------------|-----------------|
| BH1     | DFT-derived | $3.3 \pm 0.2$  | $5.3 \pm 0.4$  | $0.96 \pm 0.01$ | $0.88 \pm 0.01$ |
|         | SOAP        | $4.9 \pm 0.3$  | $7.3 \pm 0.5$  | $0.93 \pm 0.01$ | $0.84 \pm 0.01$ |
|         | PhysChem    | $6.0 \pm 0.3$  | $8.8 \pm 0.5$  | $0.90 \pm 0.01$ | $0.81 \pm 0.01$ |
|         | MFP         | $4.5 \pm 0.2$  | $6.9 \pm 0.5$  | $0.94 \pm 0.01$ | $0.85 \pm 0.01$ |
|         | OHE         | $6.4 \pm 0.3$  | $9.3 \pm 0.5$  | $0.88 \pm 0.01$ | $0.80 \pm 0.01$ |
| BH2     | DFT-derived | $13.7 \pm 0.5$ | $19.1 \pm 0.8$ | $0.69 \pm 0.02$ | $0.67 \pm 0.01$ |
|         | SOAP        | $12.8 \pm 0.5$ | $18.2 \pm 0.8$ | $0.72 \pm 0.02$ | $0.68 \pm 0.01$ |
|         | PhysChem    | $13.9 \pm 0.5$ | $19.5 \pm 0.8$ | $0.67 \pm 0.02$ | $0.66 \pm 0.01$ |
|         | MFP         | $12.1 \pm 0.5$ | $17.9 \pm 0.8$ | $0.73 \pm 0.02$ | $0.68 \pm 0.01$ |
|         | OHE         | $15.3 \pm 0.6$ | $21.9 \pm 0.9$ | $0.59 \pm 0.03$ | $0.61 \pm 0.02$ |
| SM      | DFT-derived | $7.2 \pm 0.2$  | $10.7 \pm 0.4$ | $0.86 \pm 0.01$ | $0.77 \pm 0.01$ |
|         | SOAP        | $7.5 \pm 0.2$  | $11.1 \pm 0.3$ | $0.85 \pm 0.01$ | $0.76 \pm 0.01$ |
|         | PhysChem    | $7.4 \pm 0.2$  | $10.9 \pm 0.3$ | $0.86 \pm 0.01$ | $0.77 \pm 0.01$ |
|         | MFP         | $7.4 \pm 0.2$  | $11.0 \pm 0.3$ | $0.85 \pm 0.01$ | $0.76 \pm 0.01$ |
|         | OHE         | $9.4 \pm 0.2$  | $13.1 \pm 0.3$ | $0.78 \pm 0.01$ | $0.71 \pm 0.01$ |
| SL1     | DFT-derived | $17.7 \pm 1.7$ | $33.7 \pm 4.8$ | $0.78 \pm 0.06$ | $0.50 \pm 0.03$ |
|         | SOAP        | $17.8 \pm 1.6$ | $33.4 \pm 4.4$ | $0.79 \pm 0.05$ | $0.50 \pm 0.04$ |
|         | PhysChem    | $18.4 \pm 1.8$ | $34.7 \pm 5.4$ | $0.77 \pm 0.05$ | $0.49 \pm 0.04$ |
|         | MFP         | $18.0 \pm 1.8$ | $35.1 \pm 5.1$ | $0.77 \pm 0.05$ | $0.51 \pm 0.04$ |
|         | OHE         | $22.0 \pm 2.8$ | $45.9 \pm 7.6$ | $0.61 \pm 0.10$ | $0.48 \pm 0.04$ |



**Table S4:** 5x5-fold cross-validation performance of LightGBM, KNN, and Ridge regression models trained on MFP across four datasets (BH1, BH2, SM, and SL1). Reported metrics: mean absolute error (MAE), root-mean-square error (RMSE), coefficient of determination ( $R^2$ ), and Kendall's  $\tau$ . For BH1, BH2, and SM, MAE and RMSE are in % (yields); for SL1 they reflect LC–MS product ratios.

| Dataset | Model    | MAE            | RMSE           | $R^2$           | Kendall $\tau$  |
|---------|----------|----------------|----------------|-----------------|-----------------|
| BH1     | RF       | $4.5 \pm 0.2$  | $6.9 \pm 0.5$  | $0.94 \pm 0.01$ | $0.85 \pm 0.01$ |
|         | LightGBM | $4.3 \pm 0.2$  | $6.1 \pm 0.3$  | $0.95 \pm 0.01$ | $0.86 \pm 0.01$ |
|         | KNN      | $13.0 \pm 0.4$ | $17.1 \pm 0.6$ | $0.60 \pm 0.03$ | $0.58 \pm 0.02$ |
|         | Ridge    | $12.0 \pm 0.2$ | $15.0 \pm 0.3$ | $0.70 \pm 0.02$ | $0.67 \pm 0.01$ |
| BH2     | RF       | $12.1 \pm 0.5$ | $17.9 \pm 0.8$ | $0.73 \pm 0.02$ | $0.68 \pm 0.01$ |
|         | LightGBM | $13.1 \pm 0.5$ | $18.4 \pm 0.8$ | $0.71 \pm 0.03$ | $0.65 \pm 0.02$ |
|         | KNN      | $24.0 \pm 0.5$ | $30.4 \pm 0.6$ | $0.21 \pm 0.03$ | $0.37 \pm 0.02$ |
|         | Ridge    | $23.1 \pm 0.6$ | $28.8 \pm 0.8$ | $0.29 \pm 0.03$ | $0.45 \pm 0.02$ |
| SM      | RF       | $7.4 \pm 0.2$  | $11.0 \pm 0.3$ | $0.85 \pm 0.01$ | $0.76 \pm 0.01$ |
|         | LightGBM | $7.4 \pm 0.2$  | $10.7 \pm 0.3$ | $0.86 \pm 0.01$ | $0.76 \pm 0.01$ |
|         | KNN      | $13.4 \pm 0.4$ | $17.8 \pm 0.5$ | $0.62 \pm 0.02$ | $0.57 \pm 0.01$ |
|         | Ridge    | $16.6 \pm 0.3$ | $20.0 \pm 0.4$ | $0.51 \pm 0.02$ | $0.52 \pm 0.01$ |
| SL1     | RF       | $18.0 \pm 1.8$ | $35.1 \pm 5.1$ | $0.77 \pm 0.05$ | $0.51 \pm 0.04$ |
|         | LightGBM | $18.6 \pm 1.8$ | $32.9 \pm 4.0$ | $0.79 \pm 0.06$ | $0.47 \pm 0.03$ |
|         | KNN      | $20.7 \pm 2.9$ | $42.7 \pm 7.6$ | $0.66 \pm 0.09$ | $0.47 \pm 0.03$ |
|         | Ridge    | $28.3 \pm 2.2$ | $48.5 \pm 5.5$ | $0.56 \pm 0.08$ | $0.48 \pm 0.03$ |



# Thank You

25 September 2026